TSI Instruments Ltd
Stirling Rd Cressex Business Pk
High Wycombe
Buckinghamshire HP12 3ST
UK

Service Report
For Request 300364863

Date Received: 15 Aug 2021

Date Completed: 18 Aug 2021

Sold-To Customer: 5509601
EBME Solutions Ltd
Kemp House
152-160 City Road
London
Greater London
EC1V 2NX
United Kingdom

Ship-To Customer: 5509601
Ashwin Andiapen
EBME Solutions Ltd
Kemp House
152-160 City Road
London
Greater London
EC1V 2NX
United Kingdom

Customer PO: 5100056
Service Performed: CL-4040 Calibration of Flowmeter 4040
Model Serviced: 4040 Flowmeter, 300 L/min, Air, O2, N2, 22mm
Serial Number: 40401637005

CUSTOMER RETURN REASON:
Calibrate

CUSTOMER RETURN REASON CONFIRMED: YES

AS FOUND RESULT: PASS

TECHNICIAN:
Emile Pietersen / Mahboob Hussain

TECHNICIAN'S FINDINGS:
No findings observed.

TECHNICIAN'S ACTIONS:
The following action items were taken to ensure proper operation:
The unit was cleaned, calibrated and a complete operational checkout was performed.

TSI STANDARD TASKS PERFORMED:
Through our quality policy we ensure that the best services are available to our customers. Below is a checklist of standard tasks that are performed on your instrument:

- ☒ As-Found calibration
- ☒ Clean/replace mesh screen & o-rings
- ☒ End-to-end leak check
- ☒ As-Left calibration with NIST/UKAS traceable calibration, including performance evaluation against proprietary TSI calibration bench
- ☒ Clean flow sensor & thermistor
- ☒ Upgrade firmware (if applicable)
- ☒ Clean inlets, flowbody & cover

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TSI QUALITY COMMITMENT

TSI VISION

Creating a better world by helping protect people, products and the environment, and by optimizing research and industrial processes.

TSI MISSION

TSI provides trusted measurement, application guidance, and data analytics solutions that enable our customers to make informed decisions.

OUR QUALITY POLICY

The employees of TSI are committed to meet and, where possible, exceed our customer's requirements for application guidance, precision measurement solutions and data management through the continual improvement of our products and processess.

Thank you for the opportunity to service your instrument.

Mass Flowmeter Calibration Certificate

Model: 4040
Serial Number: 40401637005
Verification date: 18-Aug-2021

Rev: H

Summary Status	
☐ As-Found	☒ In Tolerance
☒ As-Left	☐ Out of Tolerance

Environmental Conditions
Pressure: 99.9 kPa
Temperature: 20.0°C

Air Flow

Tolerance: ±(1.75% of reading or 0.050 SLPM*)

Reference (SLPM)	Measured (SLPM)	Allowable Range Min	Allowable Range Max
0.545	0.540	0.495	0.595
2.058	2.050	2.008	2.108
5.951	5.926	5.847	6.055
14.52	14.47	14.26	14.77
29.07	28.96	28.56	29.58
45.63	45.59	44.83	46.43
61.60	61.58	60.52	62.68
80.29	80.26	78.89	81.70
100.2	100.3	98.46	102.0
110.0	110.0	108.1	111.9
140.4	140.4	138.0	142.9
180.6	180.6	177.4	183.8
220.8	220.7	216.9	224.6
259.6	259.5	255.0	264.1
299.9	299.9	294.7	305.2
PASS			

Oxygen Flow

Tolerance: ±(1.75% of reading or 0.050 SLPM*)

Reference (SLPM)	Measured (SLPM)	Allowable Range Min	Allowable Range Max
0.555	0.550	0.505	0.605
2.058	2.058	2.008	2.108
6.211	6.208	6.102	6.320
14.51	14.49	14.26	14.77
29.10	29.01	28.59	29.61
45.46	45.46	44.66	46.25
61.55	61.57	60.47	62.62
80.03	80.03	78.63	81.43
100.2	100.5	98.40	101.9
110.2	110.1	108.2	112.1
139.5	139.5	137.1	141.9
179.4	179.4	176.3	182.6
219.6	219.6	215.8	223.5
260.0	260.0	255.5	264.6
300.3	300.3	295.1	305.6
PASS			

Temperature

Tolerance: ±1.000 °C

Reference (°C)	Measured (°C)	Allowable Range Min	Allowable Range Max
19.95	19.99	18.95	20.95
PASS			

Pressure

Tolerance: ±0.110 psia

Reference (psia)	Measured (psia)	Allowable Range Min	Allowable Range Max
14.49	14.49	14.38	14.60
21.45	21.46	21.34	21.56
PASS			

Internal Calibration Reference(s)

Measurement Type	Reference (FC12)	Due for Calibration
Flow	UKFC12	31-Aug-2021
Pressure	UKFC12P	31-Aug-2021
Temperature	UKFC12T	31-Aug-2021

TSI Std Conditions: 70 °F (21.11 °C) and 14.7 psia

Verified by:

TSI Instruments Ltd.
Stirling Road
High Wycombe, Bucks HP12 3ST UK

*Tolerance specified: whichever is greater

TSI does hereby certify that this flowmeter has been calibrated using TSI procedure 10000021269. The calibration of the reference standards maintain national laboratory traceability to members of the European co-operation for Accreditation (EA).

Mass Flowmeter Calibration Certificate

Model: 4040
Serial Number: 40401637005
As Found date: 17-Aug-2021

Rev: H

Summary Status	
☒ As-Found	☒ In Tolerance
☐ As-Left	☐ Out of Tolerance

Environmental Conditions
Pressure: 100.1 kPa
Temperature: 20.4°C

Air Flow

Tolerance: ±(2.00% of reading or 0.050 SLPM*)

Reference (SLPM)	Measured (SLPM)	Allowable Range Min	Allowable Range Max
0.565	0.562	0.515	0.615
2.076	2.080	2.026	2.126
5.906	5.890	5.788	6.024
14.50	14.47	14.21	14.79
29.08	28.96	28.50	29.66
45.46	45.43	44.55	46.37
61.65	61.56	60.42	62.88
80.25	80.11	78.65	81.85
100.4	100.6	98.40	102.4
110.2	110.0	108.0	112.4
139.8	139.6	137.0	142.6
180.1	179.7	176.5	183.7
219.9	219.1	215.5	224.3
259.5	258.4	254.3	264.7
300.8	299.2	294.8	306.8
PASS			

Oxygen Flow

Tolerance: ±(2.00% of reading or 0.050 SLPM*)

Reference (SLPM)	Measured (SLPM)	Allowable Range Min	Allowable Range Max
0.554	0.550	0.504	0.604
2.060	2.050	2.010	2.110
5.961	5.950	5.842	6.080
14.48	14.45	14.19	14.77
28.95	28.81	28.37	29.53
45.63	45.54	44.72	46.54
61.34	61.27	60.11	62.57
79.83	79.63	78.23	81.43
100.2	99.88	98.20	102.2
110.4	110.0	108.2	112.6
139.6	139.1	136.8	142.4
180.5	179.8	176.9	184.1
220.3	219.2	215.9	224.7
260.9	259.5	255.7	266.1
301.7	299.7	295.7	307.7
PASS			

Temperature

Tolerance: ±1.000 °C

Reference (°C)	Measured (°C)	Allowable Range Min	Allowable Range Max
20.31	20.31	19.31	21.31
PASS			

Pressure

Tolerance: ±0.145 psia

Reference (psia)	Measured (psia)	Allowable Range Min	Allowable Range Max
14.52	14.52	14.38	14.67
22.75	22.77	22.61	22.90
PASS			

Internal Calibration Reference(s)

Measurement Type	Reference (FC10)	Due for Calibration
Flow	UKFC10	31-Aug-2021
Temperature	UKFC10T	31-Aug-2021
Pressure	UKFC10P	31-Aug-2021

TSI Std Conditions: 70 °F (21.11 °C) and 14.7 psia

Verified by:

TSI Instruments Ltd.
Stirling Road
High Wycombe, Bucks HP12 3ST UK

*Tolerance specified: whichever is greater

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